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Model-based reinforcement learning for active flow control: escalating performance and accelerated training

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ABSTRACT

Active flow control (AFC) technology is crucial for achieving predominant drag reduction in marine engineering. To tackle the issue of substantial time consumption inherent in reinforcement learning (RL)-based AFC, this study presents a model-based reinforcement learning (MBRL) approach for active flow control around a circular cylinder. The surrogate models covering Re = 100–1000 under different conditions are constructed to enhance the control strategy and reduce training time, replacing the traditional computational fluid dynamics (CFD)-based RL environment. The MBRL-based strategy achieved drag reduction rates of 14.6%, 25.4%, 31.2%, and 21.8% at Re of 100, 200, 500, and 1000 respectively, demonstrating the effectiveness of the model-based reinforcement learning method in active flow control. Considering the data scarcity and complexity in predicting high Reynolds number flow fields, a transfer training strategy is implemented. Upon retraining and fine-tuning at Re = 1000, the drag reduction increased to 36.2%. Meanwhile, the proposed method achieves significantly faster policy optimisation compared to single-thread implementations, reducing the time required for convergence by orders of magnitude. This study advances the application of deep reinforcement learning in fluid dynamics by establishing a model-based framework. The MBRL method effectively adapts to complex flow field changes, facilitates flow performance optimisation, and holds promise for future implementation of active flow control technology in diverse and intricate fluid flow environments.

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KEYWORDS
Model-based reinforcement learning; active flow control; drag reduction; flow over cylinder

1. Introduction

Drag reduction is a crucial topic in the field of fluid mechanics and marine engineering, with profound implications for the design and operation of marine vessels and structures. Various techniques have been proposed to achieve significant drag-reduction performance, including passive and active flow control techniques. For passive flow control, a variety of devices such as vortex generators, fins, and porous appendages have been shown to effectively alter the wake dynamics and reduce drag (Li et al., 2023; Ünal & Gören, 2011). Zhang et al. (2024) showed that a rear-mounted porous plate can significantly reduce drag and flow-induced noise in high-Reynolds-number cylindrical flows. Compared with passive strategies, active flow control (AFC) technology has the potential to further reduce drag during transportation (Brunton & Noack, 2015). AFC systems can be categorised into open-loop and closed-loop flow control strategies based on the presence of a feedback mechanism. Unlike open-loop flow control, closed-loop AFC systems can dynamically adjust their response to real-time changes in the marine environment by incorporating feedback mechanisms, such as variations in sea state or vessel speed. Techniques such as wall oscillation and synthetic jets have shown promise in controlling flow separation and reducing drag forces in marine applications (Guastoni et al., 2023; Zhao et al., 2024). However, due to the inherent nonlinearity of the Navier-Stokes equations and the high dimensionality of the control space, finding an efficient algorithm capable of executing these actions remains a challenge. Recently, deep reinforcement learning (DRL) has demonstrated significant potential in addressing highly nonlinear and multiscale problems, making it a promising approach for tackling the challenges of AFC (Mei et al., 2021; Noack, 2023; Song et al., 2024; Wang et al., 2024; Wen et al., 2024; Yan et al., 2023).

DRL integrates the advanced representation learning capabilities of deep learning with the decision optimisation framework of reinforcement learning. It plays a

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pivotal role in the decision-making processes of agents operating in complex environments (Hachem et al., 2021), particularly within the realm of AFC. DRL allows agents to learn how to perform actions through interactions with the environment to maximise long-term cumulative rewards. Instead of receiving a pre-labelled correct answer from the environment, agents update their neural networks based on immediate or delayed rewards associated with their actions. In greater detail, agents interact with simulated environments, accumulating vast amounts of experiential data that are subsequently used to train deep neural networks. This process enables agents to execute the most effective actions based on their current states. Rabault et al. (2019) pioneered the application of DRL to train an artificial neural network for performing AFC. By utilising the synthetic jet method, a drag reduction effect of 8% was numerically achieved for the flow passing over a cylinder. Additionally, Li and Zhang (2022) employed physical information regarding flow characteristics to aid in the design of reinforcement learning (RL)-based control strategies. The reward function, which is embedded with flow stability information, contributes to enhanced stability of the controlled flow. Yan et al. (2023) successfully suppressed the vortex-induced vibration of square columns by implementing AFC methods derived from DRL. Furthermore, Fan et al. (2020) applied DRL to experimental fluid mechanics, discovering active control strategies to diminish cylindrical drag. Through numerical simulation, they gained a deeper understanding of the physical mechanisms underlying the process of drag reduction. This finding further underscores the effectiveness of DRL in the field of experimental fluid mechanics. The recent review by Vignon et al. (2023) emphasises the growing role of DRL in flow control, especially in addressing high-dimensional nonlinear dynamics. The aforementioned research indicates that the integration of DRL and AFC facilitates autonomous control and optimisation of intelligent fluid systems.

Despite these promising demonstrations of DRL-based AFC, making such approaches practical for complex and realistic flow configurations requires addressing additional challenges beyond nonlinearity and multiscale dynamics. One central difficulty is the curse of dimensionality associated with distributed-input distributed-output (DIDO) setups, which are common in fluid mechanics. Multi-agent reinforcement learning (MARL) has emerged as an important strategy: by decomposing the global control problem into multiple local agents with low-dimensional observations and actions, MARL alleviates this dimensionality bottleneck. The pioneering study of Belus et al. (2019) offered one of the first systematic analyses of DIDO formulations in this context and showed that MARL can effectively mitigate the resulting dimensionality challenges. These insights were later corroborated and extended to other flow-control configurations by Vignon et al. (2023). Building on these ideas, Font et al. (2025) successfully applied MARL in turbulent-flow applications in the SmartSOD2D framework, which reduces the area of a turbulent separation bubble at one of the highest Reynolds numbers controlled by DRL to date. Complementary to this problem decomposition at the control level, a second strand of work targets the expressiveness of the function approximator itself by redesigning the neural-network architectures used in DRL agents. Rather than relying on generic fully connected or convolutional architectures, recent studies explicitly encode physical invariants and symmetries of the flow (e.g. translational, reflectional or rotational) through invariant or equivariant architectures, thereby reducing the effective number of degrees of freedom and improving sample efficiency. Symmetry-aware network designs, as exemplified by the work of Kurz et al. (2025), have been shown to yield substantial performance gains in DRL-based flow-control tasks.

Moreover, in the practical application of DRL within the domain of AFC, a significant challenge does not arise directly from the training process of artificial neural networks but rather stems from the inherent high computational cost and lengthy simulation times associated with computational fluid dynamics (CFD). To tackle this challenge, several researchers have proposed various strategies to alleviate it (Dong et al., 2023; Ribeiro et al., 2008). For instance, Rabault and Kuhnle (2019) adopted an innovative approach by deploying multiple parallel and independent CFD models to expedite the collection of empirical data, thereby enhancing the speed of policy learning. Wang et al. (2022) employed transfer training techniques to retrain policies derived from low Reynolds number (Re) environments for use in high-Re scenarios. Furthermore, for higher-dimensional flows, Font et al. (2025) developed the SmartSOD2D framework, which employs a multi-agent reinforcement learning approach to reduce the area of turbulent separation bubbles by 9.0%. It demonstrates a successful application of DRL-based control at one of the highest Reynolds numbers to date in a completely turbulent flow. This effectively leverages prior knowledge to improve the adaptability of policy networks across diverse conditions. Despite the substantial progress that has been made in reducing training costs and enhancing the handling of complex control tasks, the fundamental issue of how to substantively decrease computation time during simulations of complex fluid environments has not been addressed so far.

This constrains agents' ability to explore and learn optimal strategies comprehensively. Current work predominantly focuses on improving efficiency through accelerating data acquisition processes or policy transfer methods rather than directly tackling the intrinsic bottleneck of slow CFD computations. This underscores the necessity for further research into more efficient solutions in this area.

While traditional model reduction approaches such as POD-based methods (Bergmann et al., 2005) have demonstrated early success in flow control through reduced-order modelling, the emergence of deep-learning-based surrogate models has significantly extended this paradigm. By constructing efficient and realistic approximate models of the flow, such surrogates can effectively replace a large fraction of the costly CFD simulations in many tasks (Du et al., 2021; Du et al., 2022; Sekar et al., 2019; Xu et al., 2023; Zhong et al., 2023). Lyu et al. (2025) further proposed a hybrid neural-network FSI solver for two tandem cylinders, coupling a deep fluid model with structural dynamics to predict vortex-induced vibrations several orders of magnitude faster than traditional simulations. These modern surrogates share conceptual parallels with model-based reinforcement learning and the 'world models' framework in DRL, where an internal model of the environment is learned and then exploited for planning and policy optimisation (Ha & Schmidhuber, 2018; Matsuo et al., 2022). Such ideas have a long history in control and robotics, and, in fluid mechanics, world-model-like approaches have already been explored in the context of deep-RL-based active flow control. For instance, Dong et al. (2023) trained surrogate models using experimental data and coupled them with an agent for active control, which effectively reduced lift fluctuations. More broadly, constructing a static surrogate model from pre-collected data also aligns with concepts in offline reinforcement learning (Levine et al., 2020; Prudencio et al., 2024). Despite their conceptual appeal, however, such world-model-type pipelines are still relatively rare in practical AFC, partly because building, validating, and maintaining the coupled surrogate-RL framework introduces substantial additional complexity compared with purely model-free DRL. Meanwhile, low-Reynolds-number flows (e.g. Re = 100) provide a controllable environment for the development and benchmarking of AFC algorithms, whereas extending the same strategies to higher-Re regimes (e.g. Re = 200–1000) introduces transitional vortical structures and weak turbulence that demand enhanced nonlinear modelling capacity and more robust learning schemes.

In light of the core challenge that DRL-based AFC is still limited by the prohibitive computational cost and wall-clock training time of CFD-based environments, this study aims to develop a model-based reinforcement learning (MBRL) approach for AFC (Deisenroth & Rasmussen, 2011) by integrating the surrogate model with the RL framework. A key contribution of this work is to provide a worked and openly available reference implementation of a world-model-like RL pipeline for active flow control, aiming to lower the practical barrier to adopting such approaches in the community. The remainder of this paper is organised as follows: Section 2 introduces the methodology used in this paper, including flow configuration, the DRL framework, model-free reinforcement learning (MFRL), MBRL, and so on. Section 3 presents the performance of the different RL models and their drag-reduction mechanisms. Finally, the conclusions and outlines direction for future are given in Section 4.

2. Numerical method

2.1. Flow configuration

Figure 1 shows the schematic of the flow configuration, which consists of a cylinder with a diameter D and a rectangle of size $L_x \times L_y = 22D \times 4.1D$. The centre of the cylinder is designated as the origin of the coordinates system. Similar to Ref (Wang et al., 2022), the cylinder centre is $0.05D$ below the centre line of the region to help trigger the vortex shedding. The active control of vortex shedding is realised by the flow injection and suction from the cylinder surface. Correspondingly, two mass flow inlets Γ_1 and Γ_2, with the mass flow rate of Q_1 and Q_2, are set respectively at an angle of 90° and 270° relative to the incoming flow direction, as illustrated in Figure 1. Both inlets cover an arc of 10°. The mass flow rate of two of inlets is controlled by the agent with $Q_1 + Q_2 = 0$ to ensure the elimination of any additional mass flow rate introduced by the jet. This approach simulates a self-contained environment where the fluid circulating within the cylinder remains constant without any net gain or loss of mass. By enforcing this balance, we eliminate the need for an external fluid source, thus simplifying the experimental setup and allowing for a more controlled study of fluid dynamics within the cylinder.

Assuming the fluid flow to be incompressible, the governing equations are as follows:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot (\nabla \mathbf{u}) = -\nabla p + Re^{-1} \Delta \mathbf{u} \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

Where $\mathbf{u}$ is the dimensionless velocity, p is the dimensionless pressure, and t is the dimensionless time. In the present simulation, the effect of volume force is ignored

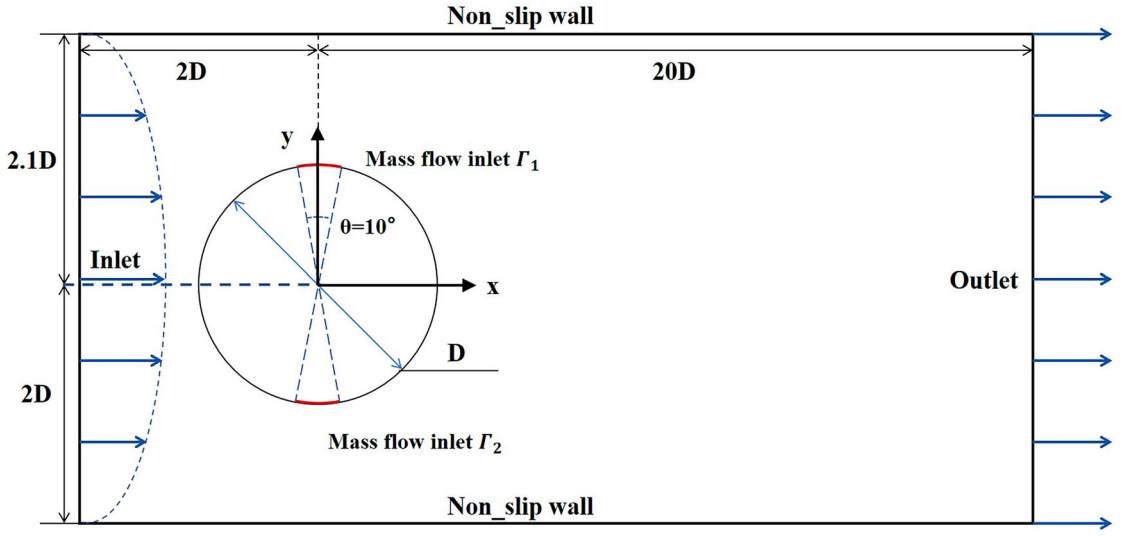


Figure 1. Schematic of the flow configuration.

because there is no density difference in the fluid. Re is the Reynolds number and defined as $Re = \overline{U}D/\upsilon$ with $\overline{U}$ the average velocity in the channel, and υ the dynamic viscosity of the fluid.

At the inlet, the parabolic profile of the stream-wise velocity $U(y)$ is given as:

$$U(y) = \frac{6(2D - y)(2D + y)}{L_y^2} \tag{3}$$

with the mean velocity $\overline{U} = 2U(0)/3 = 1$. Moreover, the outflow and non-slip boundary conditions are set at the outlet and the walls, respectively.

Figure 2 presents the grid configuration for the analyzed flow field, visually illustrating the spatial discretization strategy employed in the CFD simulation. Specifically, a structured grid is adopted in this study. Key regions, including the jet section, boundary layers, and wake regions, feature enhanced refinement to ensure an accurate representation of the flow characteristics. A transient, incompressible solver of the commercial CFD software ANSYS Fluent 23.1 is used to solve the governing equations. The coupling of velocity and pressure is handled by the SIMPLE algorithm. The second-order upwind scheme is adopted for the spatial discretization of the convective term in the momentum transport equation. Moreover, the time step is set to be 5×10^{-3} s. To evaluate the performance of AFC with different actions, the forces of drag (F_D)and lift (F_L) are computed by means of surface integrals over the cylinder surface, as follows:

$$F_D = \int (\boldsymbol{\sigma} \cdot \mathbf{n}) \cdot \mathbf{e}_x \mathrm{dS} \tag{4}$$

$$F_L = \int (\boldsymbol{\sigma} \cdot \mathbf{n}) \cdot \mathbf{e}_y \mathrm{dS} \tag{5}$$

Where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\mathbf{n}$ is the unit vector normal to the outer cylinder surface, $\mathbf{e}_x = (1,0)$, $\mathbf{e}_y = (0,1)$. Correspondingly, drag coefficient C_D and lift coefficient C_L are, respectively, calculated by

$$C_D = \frac{2F_D}{\rho \overline{U}^2 D} \tag{6}$$

$$C_L = \frac{2F_D}{\rho \overline{U}^2 D} \tag{7}$$

Based on the above method, the calculated drag coefficient of the cylinder is 3.16 and 3.47 at Re = 100 and 1000, respectively, which agrees well with the results reported in the reference (Wang et al., 2022). The Strouhal number is consistent with their results.

2.2. Framework and algorithm of DRL

In the application of DRL to achieve AFC action in the target flow passing over the cylinder, we regard the CFD simulation as a dynamic system with which the agent interacts. The DRL agent is the decision-making component in the reinforcement learning loop: it observes the state of the flow, selects the control action, and adapts its internal parameters based on the received rewards. Figure 3 illustrates the interactive process of DRL. Specifically, during the initialisation stage, a CFD model is

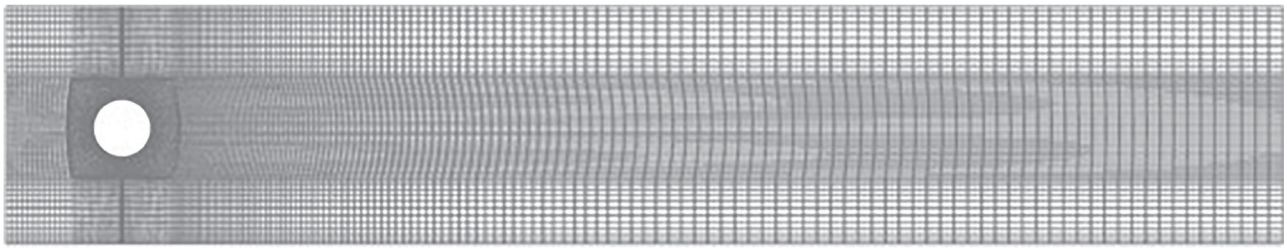


Figure 2. Structured grid configuration for the flow domain simulation.

constructed, encompassing the cylindrical geometry and its surrounding flow field, integrating variable jet velocities as controllable intervention measures by the agent. At each interaction step, the agent observes the current environmental state s_t, i.e. velocity field information obtained from the CFD model. Then, the agent selects a jet velocity as an action α_t based on the DRL policy, feeding this input into the CFD environment. The action space of the agent corresponds to the permissible range of jet velocities. It adjusts the jet velocity according to the learning objectives and strategy. Upon execution of the action α_t, the CFD simulator uses the current flow field as the initial condition and re-calculates the flow field based on the new jet velocity parameters. This process involves solving the Navier-Stokes equations to obtain the variations in unsteady flow fields, including updates to critical parameters such as velocity distribution around the cylinder, drag coefficient C_D, and lift coefficient C_L. After completing the simulation, the CFD environment feeds back the latest flow field state information to the agent, containing an assessment of the effects resulting from the agent's actions. A reward function r_t is designed to assign positive or negative rewards based on the degree of actual fluid dynamics performance improvement, e.g. reduced drag, decreased vortex strength, or optimised flow field in specific regions. Finally, using the received reward signal, the agent updates its policy weights through the DRL algorithm to better adjust the jet velocity to optimise overall flow performance in future decisions. This learning and decision-making process iterates continuously over consecutive time steps, allowing the agent to accumulate experience and refine its control strategy progressively until reaching predetermined performance metrics or satisfying training termination criteria.

With the above process, agents continuously learn to refine their strategies by observing environmental states, executing actions, and garnering immediate rewards within complex environments. This strategic optimisation process is achieved through the employment of the Proximal Policy Optimization (PPO) algorithm (see Appendix B for algorithmic details).

2.3. Model-free reinforcement learning

Figure 4 illustrates the frameworks of MFRL and MBRL. When using CFD rather than the surrogate model as the environment within DRL, we categorise it as Model-Free Reinforcement Learning (MFRL). Specifically, in both MBRL and MFRL, the agent interacts with the environment through state s_t, action a_t and reward r_t. The s_t is the information that the agent observes when interacting with the environment. The a_t represents the target jet-velocity command at the surface Γ_1. In this study, the DRL agent controls only the instantaneous jet velocity through a non-dimensional action signal a_t, which is mapped to the synthetic-jet speed. The actuator geometry (slot position and width) and the jet inclination angle (normal to the cylinder surface) are kept fixed and are not included in the optimisation space. Referring to the previous study (Wang et al., 2022), we take the array composed of 151 probes' velocity values around the cylinder at time t as s_t. With the s_t at time t as the input to the policy neural network, the neural network outputs the probability density distribution of the action and then samples the a_t according to this probability density distribution. In the continuous environment, the a_t given by the agent may be different for different states. To avoid potential numerical instability due to non-physical velocity jumps, the a_t is not directly imposed as the jet velocity at Γ_1. Instead, the actual jet velocity v_t at Γ_1 is updated according to the relaxation law. Following the benchmark methodology of Rabault et al. (2019), the velocity decay law is defined as:

$$v_{t+1} = v_t + 0.1 * (a_t - v_t) \tag{8}$$

where v_t and v_{t+1} are the jet velocities at time t and time $t+1$, respectively. This law ensures the jet velocity gradually changes from v_t to a_t in 50 time steps. Within a complete interaction process, we define an episode consisting of 80 such interactions, and during each interaction, the agent selects an action that persists for 50 time steps. As a result, a single episode encompasses a total of 4,000 time steps.

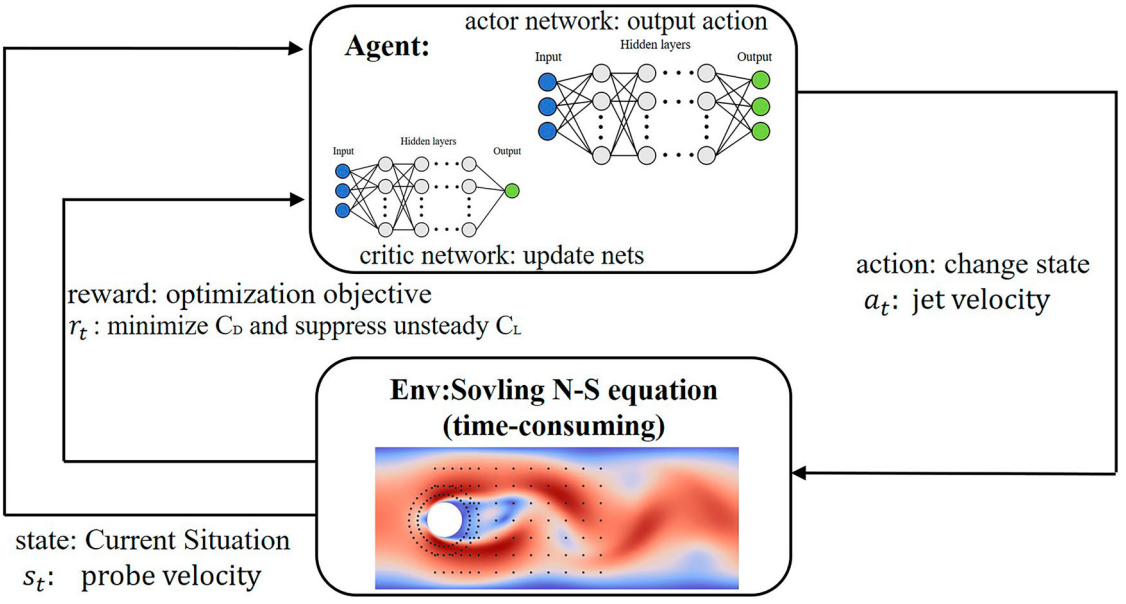


Figure 3. Interaction frame of the Agent with Environment. CFD serves as the environment of the DRL.

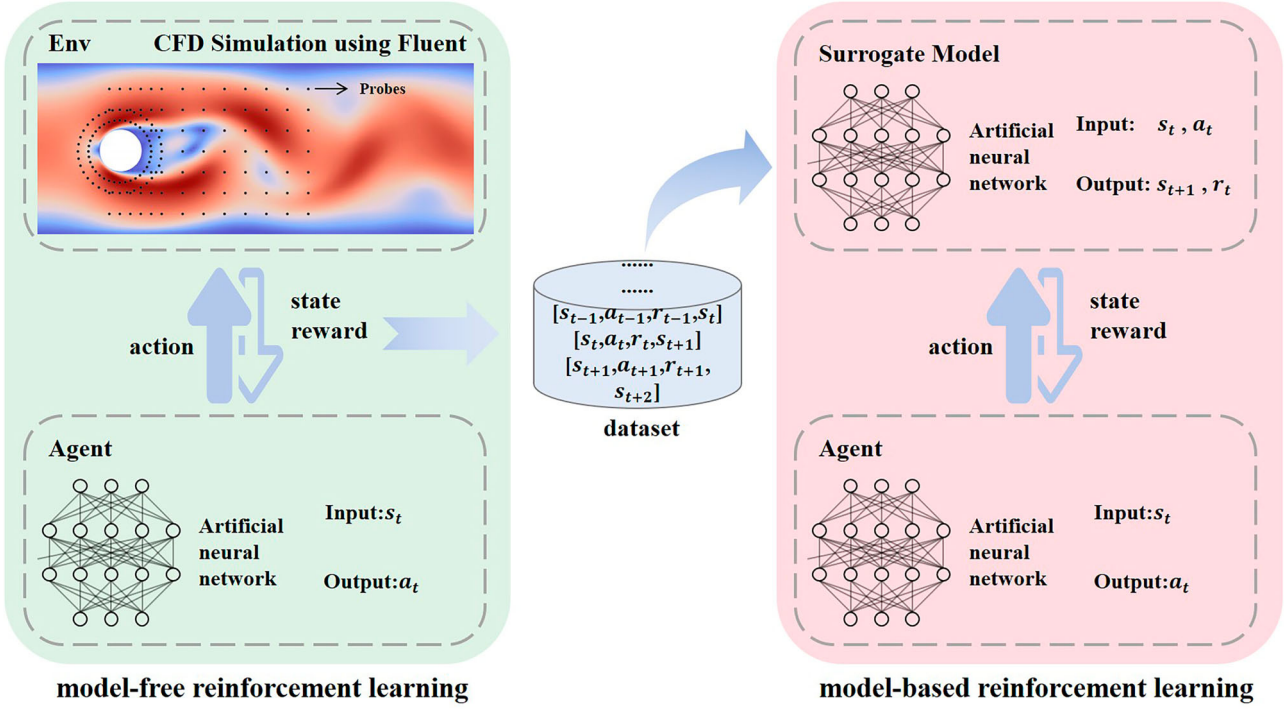


Figure 4. MFRL (left) and MBRL (right) frameworks used in this study. In MFRL, the DRL agent interacts directly with the CFD environment solved by Fluent. In MBRL, the CFD environment is replaced by a surrogate model trained from a dataset of (s_t, a_t, r_t, s_{t+1}) generated by CFD simulations.

The reward of each action is computed by the reward function. In this study, our objective is to minimise drag and lift oscillation. The reward function is formulated to simultaneously minimise drag (C_D) and suppress unsteady lift fluctuations (C_L)as originally proposed by Rabault et al. (2019):

$$r = - < C_D >_T - 0.2 < |C_L| >_T \quad (9)$$

where $< \cdot >_T$ signifies the temporally averaging over a sequence of 50 time steps.

In addition, in order to enable the Agent to recognise more complex environments and enhance its adaptability across different scenarios, there is an 80% probability that the current environment at the end of each episode will serve as the starting environment for the subsequent

episode. Conversely, there is a 20% chance that the environment will be reset back to its initial state.

2.4. Surrogate model and MBRL

In MBRL, the trained surrogate model replaces the CFD as the environment for DRL, allowing for faster interaction with the Agent, as illustrated in Figure 4. In the DRL exploration and learning process, data is accumulated to train a surrogate model. Throughout a single episode, the agent engages with the environment, generating 80 sets of tuples $< s_t, a_t, r_t, s_{t+1} >$. Upon completing 100 such episodes, a database is constructed using the 8,000 collected sets of arrays. During these interactions with the CFD environment, s_t serves as the initial field and a_t serves the boundary condition of the jet. Upon solving the Navier-Stokes equations and the reward function, the next state s_{t+1} and corresponding reward r_t are obtained. Therefore, the surrogate model is built based on the s_t and a_t to compute the next state s_{t+1} and reward r_t. Consequently, the surrogate model is trained with (s_t, a_t) as input and $(r_t, s_{t+1}\text{-}s_t)$ as output. In MBRL, given the s_t at time t, the policy network outputs a probability density distribution of actions. The a_t is selected by choosing the action with the highest probability from the probability density distribution. The surrogate model takes s_t and a_t as inputs to the policy network, which in turn generates the next state s_{t+1} and the corresponding reward r_t at time $t+1$. Thus, the surrogate model can be viewed as a mapping between (s_t, a_t) and $(r_t, s_{t+1}\text{-}s_t)$. The fast mapping of the surrogate model increases the training speed and reduces the computational cost.

Table 1 shows the parameters for the neural network used in this surrogate model. The surrogate model adopts a fully connected neural network architecture, where its input is a one-dimensional vector composed of the velocity values s_t from the 151 probes in the flow field at the current time step, coupled with the a_t taken. Similarly, the output of the model is also a one-dimensional vector, containing the predicted velocity change values $s_{t+1}\text{-}s_t$ of the 151 probes in the flow field after 50 time steps, along with the corresponding immediate r_t. The surrogate model possesses three hidden layers, which are designed to learn and capture the intricate mapping relationship between its inputs and outputs. Increasing the number of hidden layers significantly raises computational costs without yielding substantial improvements in predictive performance, while reducing the number of layers may limit the model's capacity to capture the complex mappings needed for accurate predictions, especially for high Re. Given the importance of generalisation for high Re, a three-layer architecture was selected. To prevent overfitting, we incorporated L2 regularisation (weight decay) in the optimiser with a value of 10^{-5}. This regularisation helps to constrain the model weights and improve generalisation. During the training process, we utilise Mean Squared Error (MSE) as the loss function, aiming to minimise the discrepancies between the model's predicted velocity values and the observed actual ones, as well as to estimate the immediate reward r_t as accurately as possible. This approach ultimately enhances the prediction accuracy of the surrogate model and the effectiveness of the reinforcement learning process. All surrogate models are labelled as Model_Re, where the subscript indicates the Reynolds number of the training dataset.

Table 1. The parameters for the neural network used in the surrogate model.

Parameter	Value
Input	(s_t, a_t)
Output	$(r_t, s_{t+1}\text{-}s_t)$
Hidden Layers	[304,304,304]
Learning Rate	0.0001
Batch	64
Epoch	20000
Loss Function	MSE
Optimiser	Adam

3. Results and analysis

3.1. MFRL for AFC

In this part, we first validate the effectiveness of MFRL approaches for AFC in a system at Re = 100. In the numerical experiment, a fully developed flow field was employed as the initial condition. Each episode involves an agent interacting with the environment for 80 consecutive steps, where each action remains in effect for 50 time steps. Figure 5 illustrates the evolution of rewards across various episodes during the training phase. Clearly, the reward associated with the strategy consistently increases throughout the learning process, achieving convergence at around the 150th episode.

Figure 6 presents the outcome of the Agent's interaction with the flow environment according to the adopted strategy. From the illustration, it can be discerned that in the absence of any active flow control interventions ($Q = 0$), the drag coefficient oscillates within 1.1%, displaying a cyclic pattern. Once synthetic jets are applied to the cylindrical surface, however, a significant decrease in the surface drag coefficient occurs, falling from 3.16 to 2.93, i.e. a 7.3% reduction in drag force. Additionally, the fluctuation range of the drag coefficient has been effectively curbed, shrinking from an original value of 0.03 down to 0.005. The results are close to Wang et al. (2022).

Moreover, we conduct a comparison of the detailed flow characteristics under conditions with and without

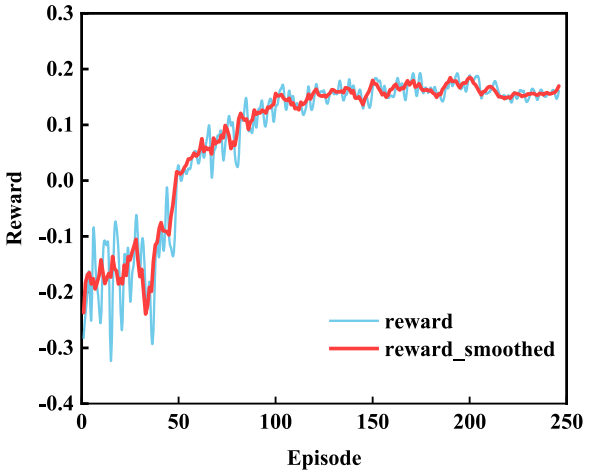


Figure 5. Reward function in the learning process at Re = 100.

control. For this purpose, the snapshots taken at the moment with the minimum drag coefficient are selected. Figure 7 shows the comparison of detailed flow characteristics, including velocity field, pressure field, and vorticity field without and with control obtained by MFRL. Compared with the case without control, the flow with control in the up-down direction alters the original pattern of unidirectional axial flow, introducing an additional radial component. This rearranges the streamline structure behind the cylinder, causing the boundary layer that would otherwise separate relatively quickly to delay its separation, thereby elongating the wake development region. Moreover, the pressure distribution in the controlled case also shows an effective reduction along the surface of the cylinder. Consequently, the drag coefficient from the pressure drop has decreased from 2.45 to 2.29, thereby contributing significantly to the decrease in shape drag, which is indeed the main source of reduced drag. As for the vorticity distribution, in the uncontrolled cylinder flow scenario, there exists a strong shear effect on the side surface of the cylinder due to the rapid change in direction forced upon the fluid. Concurrently, the downstream of the separation point, the separated fluid forms vortex structures that rapidly grow, culminating in a periodic Kármán vortex street behind the cylinder. However, with control, jets perturb the flow field and delay the formation and subsequent shedding of vortex pairs, thereby altering the vortex dynamics. The above results clearly demonstrate the efficacy of MFRL methods for AFC. Implementing AFC using MFRL methods, the addition of surface jets can extend the wake region, effectively reducing shape drag. Simultaneously, it also suppresses the formation and shedding of vortex pairs, diminishing vortex-induced drag and thereby optimising the overall performance of the flow field.

Table 2. Training and test loss of surrogate model.

	Training loss	Test loss	Relative prediction error (36 points)
Re = 100	0.0002	0.0002	1.9234%
Re = 200	0.0003	0.0003	2.0159%
Re = 500	0.0009	0.0010	0.9408%
Re = 1000	0.0017	0.0020	2.5528%

3.2. MBRL for AFC

This section focuses on constructing a surrogate model and validating the efficacy of the MBRL method. The dataset for training the surrogate model is obtained through the interactions between Agent and ANSYS Fluent. Within each interaction episode, an array is assembled featuring the current state s_t, the action a_t, the reward r_t and the next state s_{t+1}. A collection of 100 episodes has been compiled and utilised for training the surrogate model.

To comprehensively evaluate the predictive capability of the surrogate model, Figure 8 and Table 2 illustrate the training errors, test errors and predictive accuracy under different Re. Figure 8 displays the training error and test error during the surrogate model's training process. Moreover, to evaluate the reproducibility of the surrogate model, we trained the surrogate model using different random seeds, which demonstrates the good consistency and stability of these surrogate models. The associated results can be found in https://github.com/WangyanFR/MBRLforAFC. Across all surrogate models, both training and test errors converge favourably, suggesting stable learning behaviour. At Re = 100 and Re = 200, Model_100 and Model_200 achieve remarkably low training and test errors, indicating that the surrogate model attains high accuracy in capturing fluid dynamics under stable, low-Re conditions. When Re increases to 500 and 1000, the test errors of Model_500 and Model_1000 rise to approximately 0.001 and 0.002, respectively. The increase in prediction error is attributed to heightened flow instabilities and stronger nonlinear interactions. Additionally, comparisons between CFD simulation results and surrogate model predictions are conducted for 36 monitoring points around a cylinder under specified states and actions. As shown in Table 2, the average relative error between the surrogate model's predictions and CFD simulation results ranges from 0.94% to 2.55%. The predicted data generated by the surrogate model closely align with actual CFD simulation results. This demonstrates that, under identical initial conditions and control inputs, the surrogate model can effectively predict future states.

Based on the agent model established previously, we substituted the environment for DRL with Model_100

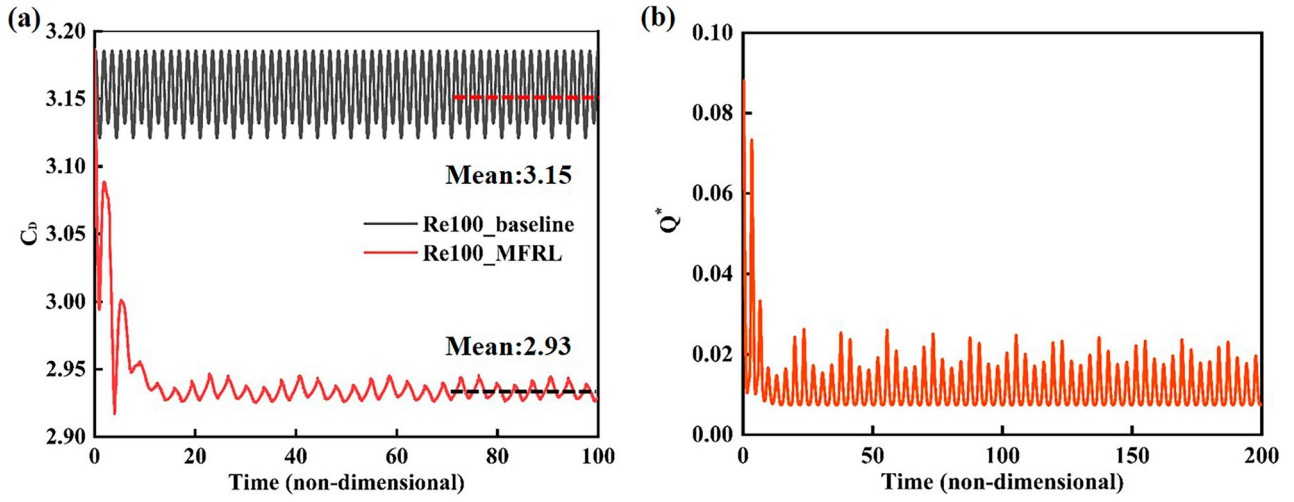


Figure 6. The impact of Agent_100_MFRL control on drag coefficient and jet velocity in a deterministic model at Re = 100: (a) comparison of drag coefficient with and without control by MFRL; (b) jet velocity obtained by Agent_100_MFRL.

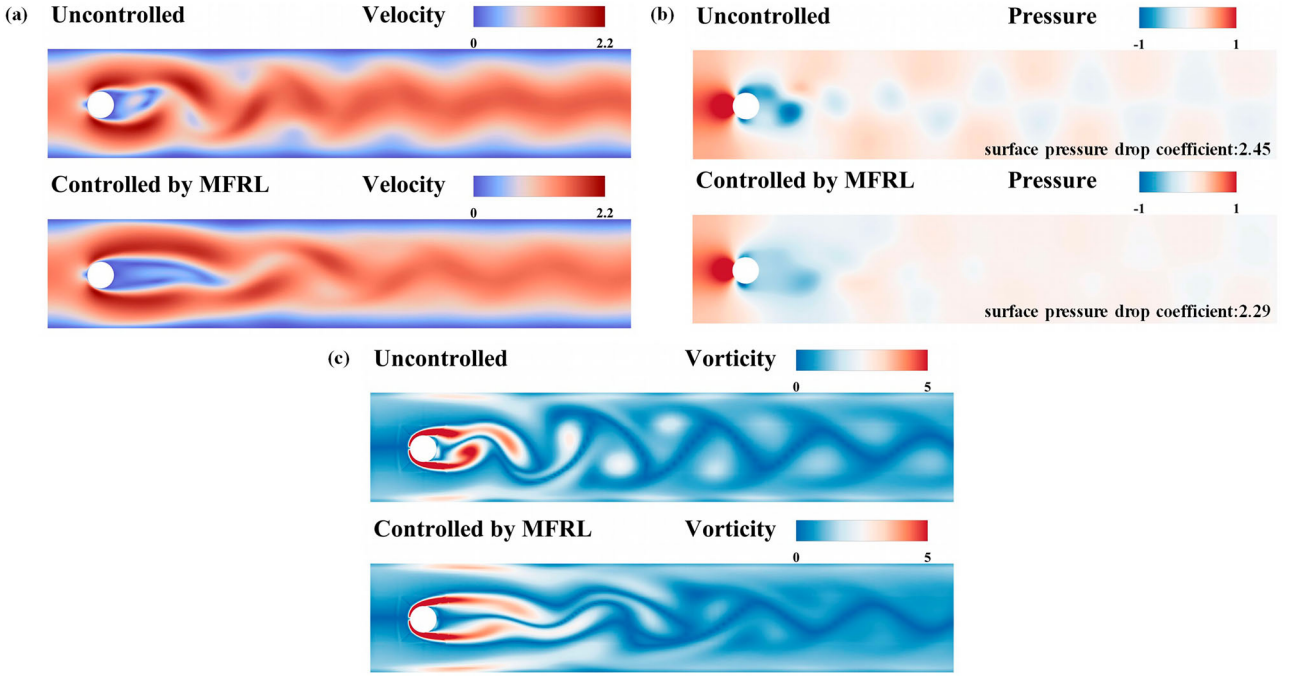


Figure 7. Comparisons of detailed flow characteristics for the cases without and with MFRL control: (a) Velocity field; (b) Pressure field; (c) Vorticity field.

(surrogate model trained over data corresponding to Re = 100). Figure 9 illustrates the variations in the drag coefficient of the flow around a cylinder under three distinct scenarios, i.e. without any control measures, under control by a MFRL method, and under control by a MBRL method. This figure allows us to compare the performance of the flow under the influence of the three control approaches. Compared with AFC governed by MFRL methods, AFC based on MBRL leads to an increased jet velocity, displaying periodic flow features. The increase in jet velocity reduces the drag coefficient by 14.6%, which is higher than the 8% drag reduction in other works (Rabault et al., 2019; Wang et al., 2022). Additionally, the amplitude of the drag coefficient oscillations is maintained within 0.2%. In other words, the application of MBRL for AFC delivers superior performance compared to that of MFRL. Furthermore, Figure 10 compares the detailed flow characteristics, including the velocity fields, pressure fields, and vorticity fields under control action obtained by Agent_100_MBRL and Agent_100_MFRL. It can be observed that both strategies exhibit similar characteristics in their velocity and vorticity fields. However,

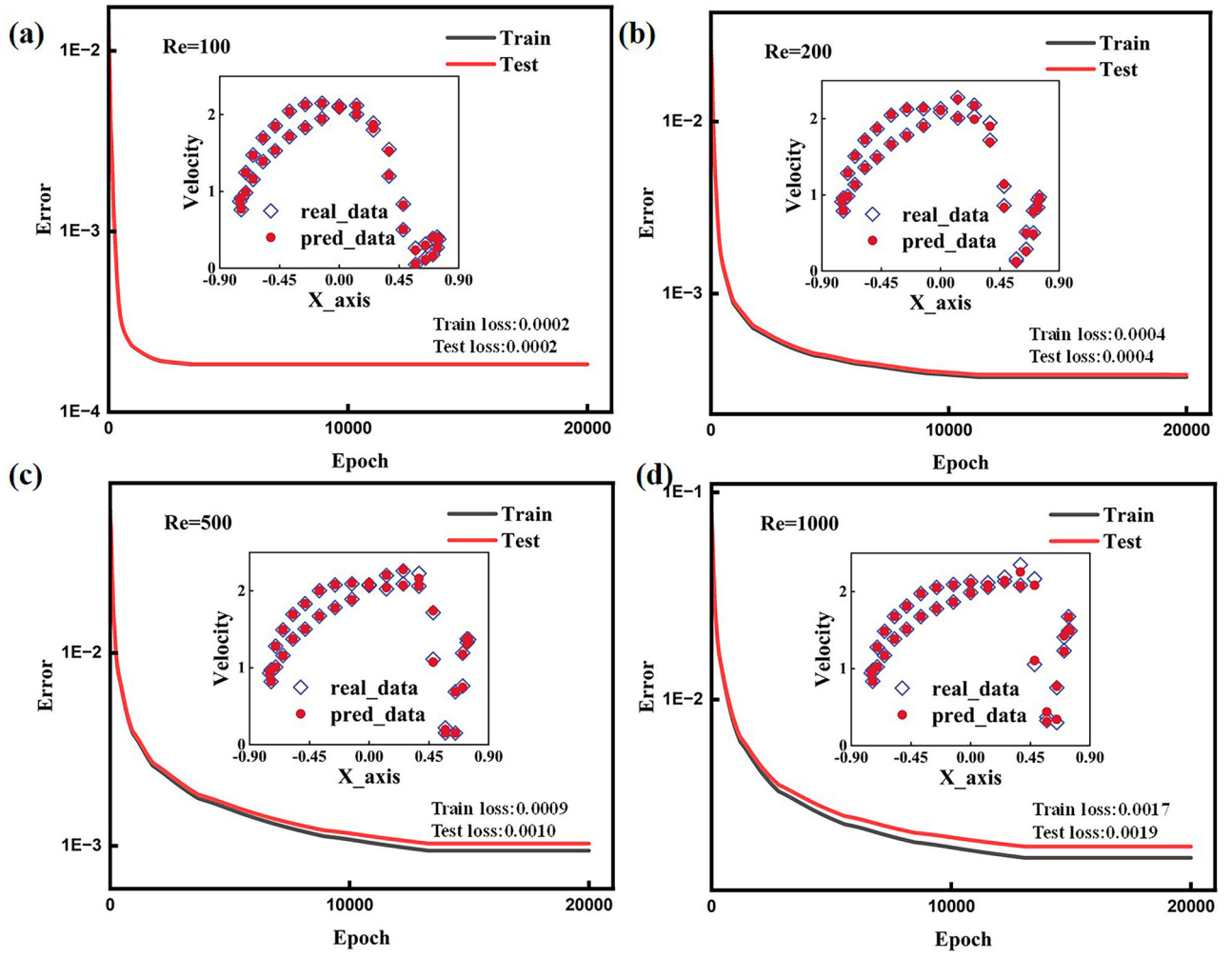


Figure 8. Train/test error comparison and probe point validation for surrogate Models: (a) Model_100; (b) Model_200; (c) Model_500; (d) Model_1000. Insets show errors between real and predicted values at select probe points within each subplot.

a clear difference appears in the surface pressure on the cylinder. As indicated by the labels in the lower-right part of Figure 10(b), the surface pressure-drop coefficient decreases from 2.29 for the MFRL-controlled case to 1.92 for the MBRL-controlled case, which directly leads to a reduction in pressure-difference drag.

The MBRL framework is then extended to tackle AFC challenges at elevated Re (Re = 200, Re = 500, and Re = 1000). The DRL environments are instantiated through Agent models designated as Model_200, Model_500, and Model_1000, each engaging in interaction with the respective agents to derive corresponding control strategies. At Re = 200, 500, and 1000, three agents were trained, respectively named Agent_200_MBRL, Agent_500_MBRL, and Agent_1000_MBRL. Figure 11 illustrates the variations in drag coefficient and jet velocity in the deterministic model for three distinct strategies. It becomes evident that the Agent successfully diminishes the drag coefficient of the flow encircling the cylinder to a significant extent. Concurrently, Figure 11 also reveals that the fluctuations in the drag coefficient have been notably curtailed. Employing MBRL methods, the cylinder achieved drag force reductions of 24.60%, 30.61%, and 22.03% at Re = 200, 500, and 1000, respectively, across varying fluid environments. As the Re increases, the complexity of the flow field significantly escalates, posing greater demands on the predictive accuracy of the surrogate model. When Re rises from 500 to 1000, the surrogate model encounters more intricate vortex structures and stronger nonlinear interactions, leading to a decrease in prediction accuracy. In complex flow environments, agents may not be able to accurately evaluate the long-term effects of certain control strategies, resulting in local optimality. This limitation constrains the surrogate model's ability to precisely evaluate and optimise control strategies, thereby affecting the global optimisation of the final drag reduction strategy and resulting in a lower drag reduction rate at

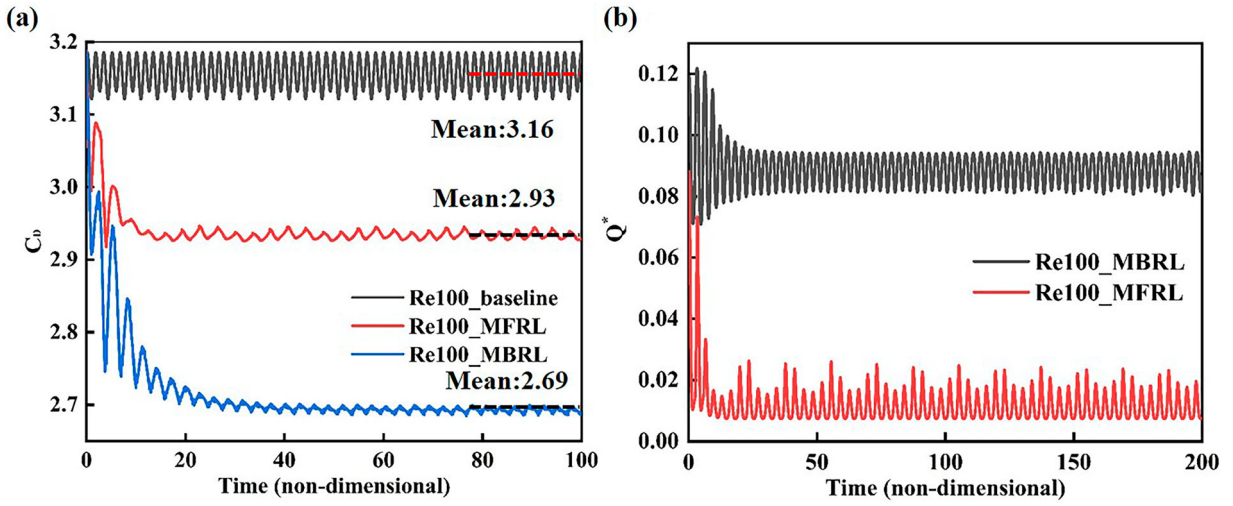


Figure 9. Control performance of Agent_100_MFRL versus Agent_100_MBRL evaluated through CFD validation: (a) Drag coefficient trajectories; (b) Jet velocity profiles. Results obtained by deploying the surrogate-trained agents in the true CFD solver at Re = 100.

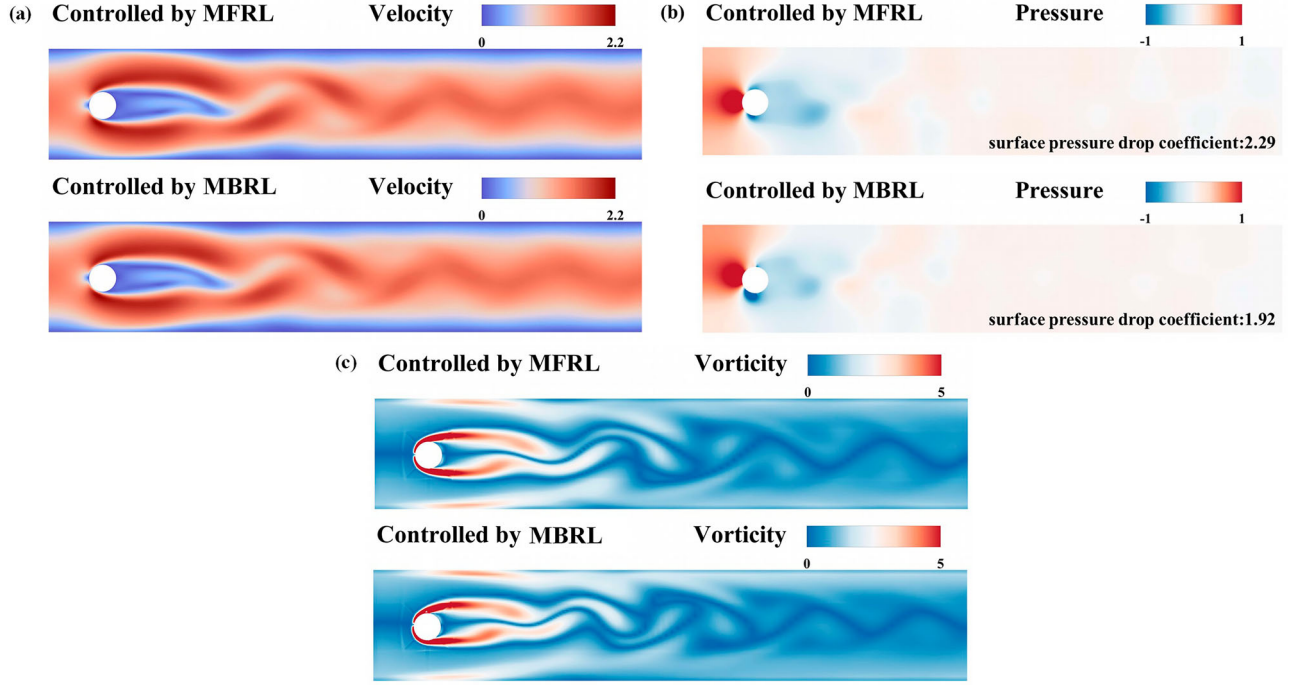


Figure 10. Comparisons on the detailed flow characteristics for the cases with controls obtained by MFRL and MBRL, respectively: (a) velocity field; (b) pressure field; (c) vorticity field.

Re = 1000 compared to Re = 500. The findings demonstrate that the MBRL methodology is capable of achieving efficacious control in nonlinear laminar wake flows at low Reynolds numbers (Re = 100–1000).

In sum, this section establishes a framework employing surrogate models and develops an MBRL method. While both MFRL and MBRL methods achieve drag reduction compared to the uncontrolled scenario, MBRL-based AFC demonstrates a more pronounced effect. Despite showing similar trends in the velocity and vorticity fields, MBRL exhibits a stronger influence on the pressure field. It results in a substantial decrease in the drag coefficient of the pressure drop on the cylinder surface, which directly translates into a more effective mitigation of shape drag. Moreover, the MBRL method is further extended to operate in environments with high-Re for AFC, where it consistently achieves a significant reduction in the drag coefficient. This substantiates the practicality of the MBRL method in the domain of AFC across a range of Re.

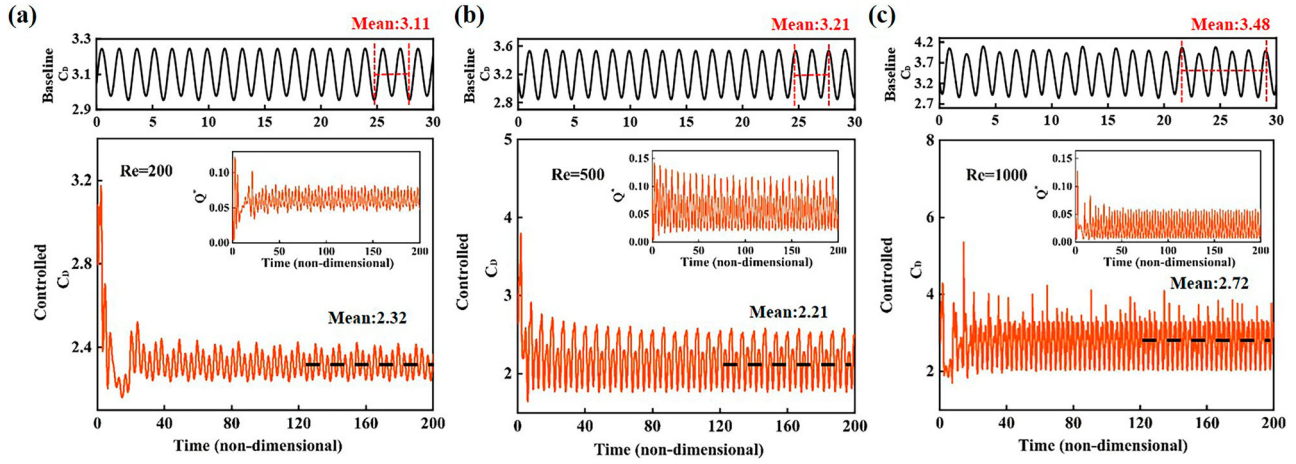


Figure 11. Comparisons on the drag coefficient for the cases without control and with control obtained by (a) Agent_200_MBRL at Re = 200; (b) Agent_500_MBRL at Re = 500; (c) Agent_1000_MBRL at Re = 1000. Inset of the figure shows the changes in jet velocity under controlled conditions.

3.3. MBTT for AFC

While the Re = 100 case serves as a laminar benchmark for control policy calibration, whether these strategies maintain efficacy when confronted with the amplified nonlinearity and multiscale vorticity inherent in higher Reynolds regimes (Re = 200–1000) is still necessary to be evaluated. This progression deliberately probes the transition from laminar coherence to weakly turbulent disorder. This serves to assess and validate the adaptability of our trained strategies to diverse flow conditions. To this end, we directly transfer the policy trained in the Re = 100 environment to the testing environments of Re = 200, Re = 500, and Re = 1000. The progression of the drag coefficient is visually depicted in Figure 12. The trained Agent exhibits a periodic pattern within the Re = 200, 500, and 1000 environments. Moreover, there is a drop in the drag coefficient, declining from 3.11, 3.21, and 3.48 to 2.76, 3.15, and 3.35, respectively. This indicates a certain degree of drag reduction compared to the uncontrolled flow around the cylinder. Nonetheless, it is noteworthy that the variation in the drag coefficient displays substantial fluctuations. The drag fluctuations are 0.34, 0.82, and 1.13 at Re = 200, 500, and 1000, respectively. These results collectively suggest that the capability of agents trained in lower Re environments to handle increasingly complex scenarios diminishes progressively. The reduced effectiveness of the low-Re-trained model (Re = 100) with classical fluid dynamics expectations at higher Reynolds numbers aligns. As Re increases, the flow exhibits more complex structures and may have different sensitivities and receptivities (Tang et al., 2020). This phenomenon is further amplified when the flow transitions from laminar to weakly turbulent regimes, consistent with prior studies (Varela et al., 2022).

Compared with ordinary DRL, the MBRL method generates learning experience in the process of strategy learning by the agent model, so the performance of the strategy is closely related to the accuracy of the model. Facing more complex environments, higher model errors may result in lower performance of learned strategies. To address this problem, we propose the following model-based transfer training (MBTT) as a means to enhance the performance of the policy. Instead of initialising the neural network's weight and bias values via the orthogonal method, the strategy involves transferring knowledge acquired under Re = 100 to more intricate environments. This is achieved by employing the parameters of the neural network pre-trained at Re = 100 as the starting point for the Agent's parameters, which subsequently engage with Model_200, Model_500, and Model_1000 in separate interactions. A deterministic test under different Re was then performed to evaluate the retrained strategy.

Figure 13 illustrates the performance of different agents trained by MBTT. The MBTT method for AFC consistently demonstrates effectiveness across all cases, effectively reducing drag. Despite some challenges during the transition to more complex environments, the MBTT method effectively harnesses prior knowledge to enhance adaptability, ultimately leading to improved drag reduction rates and efficiency in learning processes. Relative to the case without control, these strategies yield significant reductions in drag. Specifically, at Re = 200, 500, and 1000, the mean drag coefficients achieved through strategies trained with the MBTT method are respectively reduced by 24.1%, 28.7%, and 36.2%.

In comparison with the MBRL method, the MBTT method for AFC exhibits varying performance for different Re. At Re = 200, strategies trained using the MBTT method show a slight increase in the mean

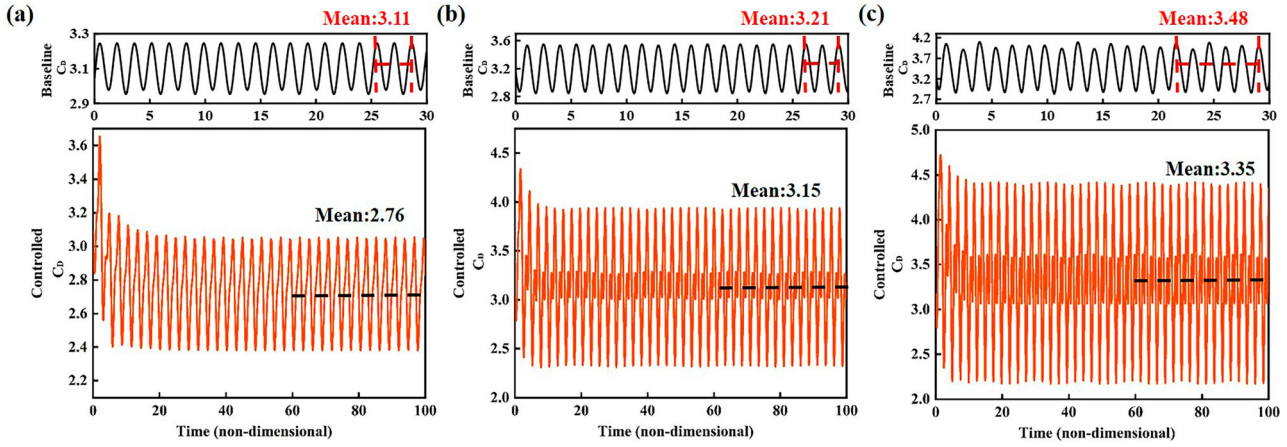


Figure 12. The comparison of the drag coefficient for a cylinder without control versus with control by Agent_100_MBRL at (a) Re = 200, (b) Re = 500 and (c) Re = 1000.

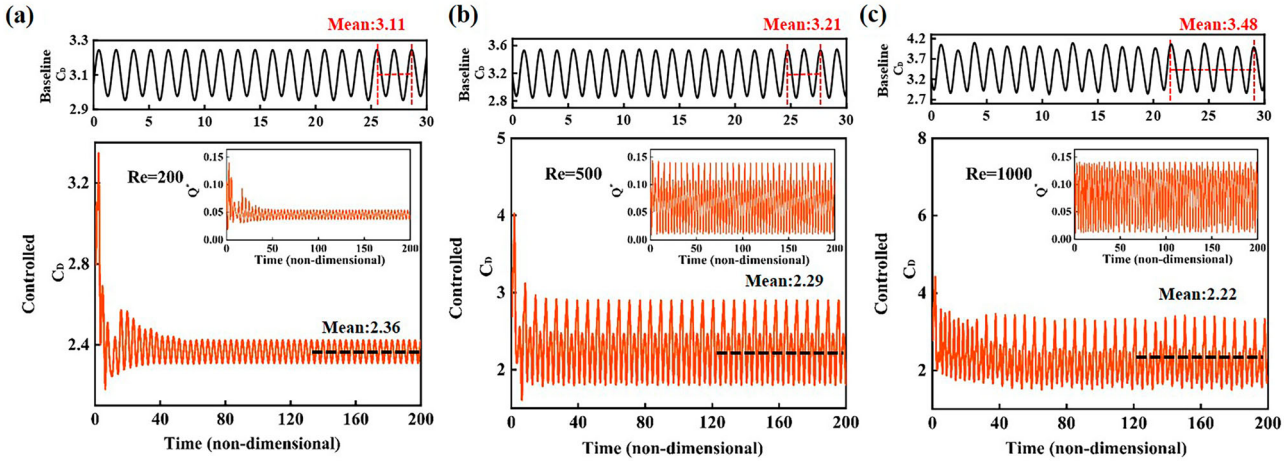


Figure 13. Comparisons on the drag coefficient for the cases without control and with control obtained by (a) Agent_200_MBTT at Re = 200; (b) Agent_500_MBTT at Re = 500; (c) Agent_1000_MBTT at Re = 1000. Inset of the figure shows the changes in jet velocity under controlled conditions.

drag coefficients, but the drag-coefficient time histories exhibit more regular and almost sinusoidal oscillations with reduced high-frequency fluctuations compared with those obtained by MBRL. At Re = 1000, the MBTT approach significantly outperforms the MBRL method, achieving a drag reduction rate of 36.2%, which is consistent with Wang et al. (2022) but accomplished in less time. In Comparison to MBRL, the amplified effectiveness of MBTT at Re = 1000 can potentially be explained in several respects. Firstly, considering that the Model_1000 might not accurately replicate the Re = 1000 environment when model error is high, the employment of prior knowledge can offset this model inaccuracy. Transfer training leverages the knowledge gained in the Re = 100 environment to inform better action choices in the Re = 1000 environment, thereby enhancing the learning process. Secondly, using the MBRL method directly in the Re = 1000 environment can lead to unreliable predictions and unstable exploration due to the high model error associated with the surrogate model in this complex scenario. This may hinder the agent from fully discovering and adapting to the specifics and nuances of the Re = 1000 environment, producing suboptimal results. By adopting the Re = 100 strategy as the foundational strategy for transfer training, the potential for agent model error can be somewhat mitigated, allowing for steadier and more beneficial exploration despite relying on an imperfect model. Lastly, direct application of the MBRL approach might not sufficiently adjust to the unique properties and dynamics within the Re = 1000 environment. With high model errors, the agents may struggle to effectively discern and respond to subtle changes and complexities within the Re = 1000 environment, which can contribute to elevated drag coefficients. On the other hand, transfer training, capitalising on previously honed

strategies and models, enables better exploitation of past experiences, which offers superior model adaptability and thereby delivers improved performance. Regarding the slight decrease in drag reduction rates at Re = 200 and Re = 500, it may be due to the negative transfer phenomenon during strategy transfer. That is, some control strategies effective at low Re may not be fully suitable for fluid characteristics at medium Re, causing certain elements in the initial strategy to become obstacles to learning in the new environment. Thus, the MBTT method serves as a complementary approach when the accuracy of the surrogate model is not ideal. It leverages existing strategic experience to provide a possible route for Agent exploration in the surrogate model.

To further understand the drag reduction mechanism of the AFC by MBTT, Figure 14 compares the detailed flow characteristics at Re = 1000 for the cases without control, with control by Agent_1000_MBRL, and with control by Agent_1000_MBTT. It can be discerned that while the differences in flow patterns between the controlled and uncontrolled states at Re = 1000 are relatively minor, the pressure coefficient differs significantly. As indicated by the labels in the lower-right part of Figure 14(b), the surface pressure-drop coefficient on the cylinder is 2.68 in the uncontrolled case, decreases to 1.82 under the control of Agent_1000_MBRL, and is further reduced to 1.28 under the control of Agent_1000_MBTT, which directly leads to a stronger reduction in pressure-difference drag.

3.4. Discussion

This section further discusses the overall performance of different AFC models from the viewpoints of drag reduction and computational efficiency. Table 3 presents a comparison of various drag reduction strategies under different Re. In order to more intuitively show the effect of different drag reduction strategies, the percentage relative to the uncontrolled drag coefficient is added after the absolute drag coefficient. While both MFRL and MBRL methods achieve reductions in the drag coefficient compared to the uncontrolled scenario, MBRL-based AFC demonstrates a more pronounced effect, reducing the drag coefficient to 2.69 (at Re = 100) – a level surpassing previous research outcomes by a considerable margin. This is due to the enhanced interaction efficiency of MBRL, which allows for the accumulation of more experience in a shorter period of time. Furthermore, in the face of complex environments, establishing an accurate surrogate model with limited samples becomes challenging. Therefore, the MBTT method is established for AFC. By utilising policies trained at low Re as initial strategies and incorporating transfer training in complex environments, we can effectively and expeditiously generate improved control strategies. At Re = 1000, the agent trained using the MBTT method achieved a drag reduction rate of 36.2%. It is noteworthy that our method, through efficient exploration in extended action spaces by the reinforcement learning agent guided by the surrogate model, achieves a 24.1% drag reduction rate at Re = 200 (Tang et al., 2020: 21.6%), and reaches 36.2% at Re = 1000 (Ren et al., 2021: 30%; Varela et al., 2022: 20%)

Moreover, it is noteworthy that CFD usually involves high-dimensional, non-linear computationally intensive simulation, so the interaction time between Agent and CFD takes a long time in the learning process of strategy. Taking the example of flow control around a cylinder in this study, surrogate models mimic the behaviour of the CFD environment enabling swift predictions within a much shorter timeframe, thereby drastically reducing policy learning timespan. For example, this work was executed on hardware equipped with an AMD Ryzen 7 4800H CPU and an NVIDIA GeForce GTX 1650 GPU. As shown in Figure 15, employing the MFRL method results in an average execution time of about 40 min per episode, whereas adopting the MBRL method reduces this duration to only 0.84 s. This substantial acceleration has greatly expedited the iterative learning process and optimises the overall efficiency of our research. Moreover, this work also operates within a somewhat simplified scenario. As the complexity escalates-such as transitioning to more intricate three-dimensional environment or higher-resolution mesh configurations – the computational burden of traditional CFD methods becomes increasingly onerous, potentially leading to a considerable decline in learning efficiency for the policies involved. At such junctures, the advantages of MBRL become even more pronounced: while maintaining high-speed learning capabilities, MBRL holds promise for extending its application to higher-dimensional and finer-grained complexities in fluid dynamics scenarios. This extension would effectively address the challenges posed by large-scale, high-fidelity simulations, thus demonstrating the potential of DRL applications in high-performance computing.

These results are encouraging, but two issues deserve further discussion: scalability to more complex flows and the integration with recent DRL advances. An important open question still concerns the scalability of the proposed model-based framework to more complex flow configurations. In the present work, we restrict attention to two-dimensional cylinder flow for Re = 100–1000, spanning laminar to weakly turbulent regimes. Within this range, our results already indicate

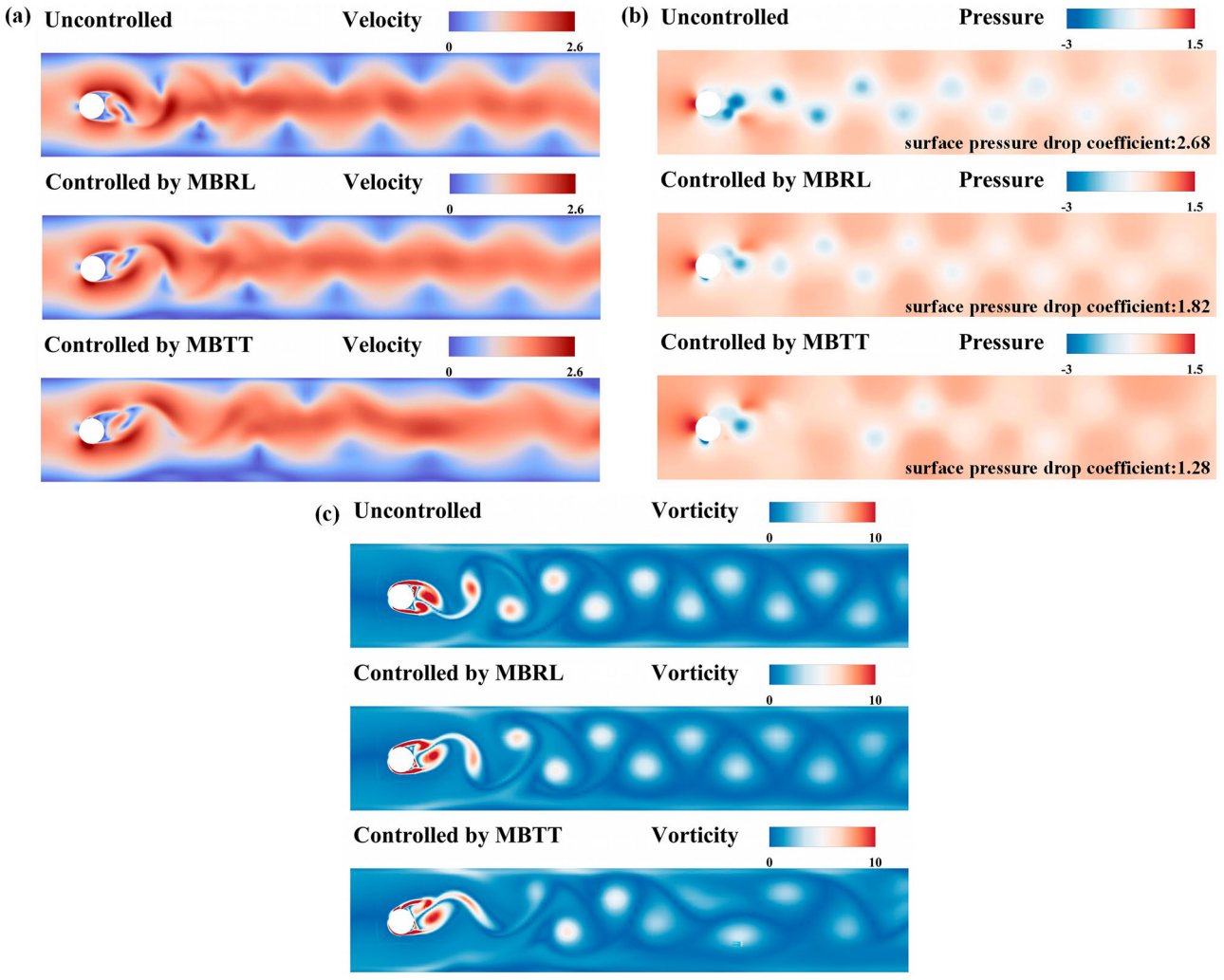


Figure 14. Comparison of flow characteristics for cases without control, with control by MBRL with control by MBTT: (a) Velocity field; (b) Pressure field; (c) Vorticity field.

Table 3. Comparison of drag reduction effect of different strategies.

	Drag drag reduction rate in deterministic control. Note: (·) is the mean C_D.			
	Re = 100	Re = 200	Re = 500	Re = 1000
Without control	3.16	3.11	3.21	3.48
Agent_100_MFRL	7.3% (2.93)	11.3% (2.76)	1.9% (3.15)	4.7% (3.35)
MBRL	14.9% (2.69)	25.4% (2.32)	31.2% (2.21)	21.8% (2.72)
MBTT	/	24.1% (2.36)	28.7% (2.29)	36.2% (2.22)
Tang et al. (2020)	5.7%	21.6%	/	/
Ren et al. (2021)	/	/	/	30%
Varela et al. (2022)	/	/	/	20%

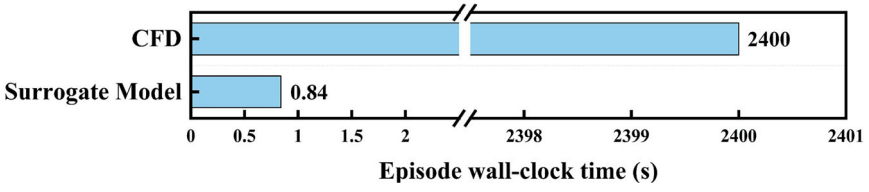


Figure 15. Comparison of episode wall-clock time between the CFD environment and the surrogate model environment.

that surrogate accuracy deteriorates as Re increases and the flow dynamics become more nonlinear and multi-scale, and that prediction errors tend to accumulate over long horizons, which can adversely affect the performance of the DRL controller. These observations suggest that a straightforward extension of the current surrogate architecture is unlikely to be sufficient for fully turbulent configurations, such as the separation-bubble flow considered by Font et al. We regard this, however, as a challenge for surrogate modelling rather than a fundamental obstacle to world-model-based DRL. In particular, recent developments in neural-operator architectures and other field-to-field surrogate models provide a promising way to learn state-to-state evolution operators directly on high-dimensional flow fields over a range of parameters. Coupling such operator-learning surrogates with multi-fidelity or physics-informed regularisation, uncertainty-aware world models, and possibly localised or multi-agent decompositions could yield the 'reasonably good' world models required to tackle high-Reynolds-number, fully turbulent separation-bubble flows. We therefore view the extension of the present framework to such configurations as a challenging but realistic direction for future work, and our current study as an intermediate step that clarifies both the potential and the bottlenecks of this class of methods.

Beyond improving the accuracy of the surrogate model itself, other recent developments in DRL-based flow control are largely orthogonal and complementary to the strategy proposed here. In particular, many realistic AFC problems naturally lead to distributed-input distributed-output formulations, for which MARL has proved an effective way to alleviate the curse of dimensionality by decomposing the global control task into interacting local agents with low-dimensional observations and actions (e.g. Belus et al., Vignon et al., Suárez et al., Font et al.). In parallel, symmetry-aware neural-network architectures that explicitly encode invariants and symmetries of the flow, such as the equivariant designs proposed by Kurz et al., have been shown to significantly improve sample efficiency and generalisation. These 'force multipliers' can in principle be combined with the present model-based framework, so that MARL and symmetry-preserving networks operate on top of a learned world model, offering a promising route towards scalable DRL controllers for high-dimensional, realistic AFC configurations.

4. Conclusions

To address the challenge of heavy time cost in RL-based Active Flow Control (AFC), this study proposes the frameworks of MBRL, which integrates the surrogate models with reinforcement learning to replace the CFD computations and achieve efficient AFC. The performance of the MBRL method is evaluated across multiple Re environments. Moreover, the MBRL is extended to MBTT, which transfers the policy learned under low Re serves to more complex environments and can realise superior AFC. All the obtained policies are tested in actual environments to assess the efficacy of both the MBRL and MBTT methods. The following conclusions are drawn:

(1) At Re $=$ 100, the adoption of the MFRL method for AFC realises a drag reduction effect of 7.3%. Further application of the MBRL method leads to an increase in drag reduction efficiency to 14.6%. This improvement in efficiency is attributed to the agent increasing the jet speed through faster agent-environment interactions. It contributes to a further reduction in shape drag.
(2) We use strategies learned at low Re to transfer to more complex environments. At Re $=$ 1000, the drag reduction efficiency is increased to 36.2%. This improvement underscores the value of incorporating model knowledge into the learning process, enabling algorithms to make more informed decisions.
(3) The MBRL and MBTT methods also expedite the learning process. It lays a solid foundation for the implementation of the AFC strategy in more complex and more extensive fluid flow environments in the future.

In summary, the proposed MBRL and MBTT frameworks provide an effective and computationally efficient pathway for RL-based active flow control across a range of Reynolds numbers. By replacing expensive CFD interactions with surrogate-guided learning and validating the learned policies in the original environments, this study demonstrates a practical step toward scalable, data-efficient AFC. We anticipate that continued advances in surrogate modelling and integration with complementary DRL techniques will further extend the applicability of this framework to higher-dimensional and more challenging flow-control problems.

Author contributions

CRediT: **Yan Wang:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Writing – original draft; **Hong-Na Zhang:** Formal analysis, Funding acquisition, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing; **Chun-Yu Wang:** Data curation, Formal analysis, Visualization; **Xin Zheng:** Formal analysis, Supervision, Writing – review & editing; **Xiao-Bin Li:**

Funding acquisition, Investigation, Resources, Supervision, Validation, Writing – review & editing; **Feng-Chen Li:** Methodology, Resources, Supervision, Writing – review & editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Data will be made available on request.

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